

Complex Numbers

1.) Show that $\operatorname{Re}(z)^2 \leq |z|^2$ and under what circumstances equality holds.

Solution: Write $z = a + bi$ such that $z^* = a - bi$. So, the real part becomes $\operatorname{Re}(z)^2 = a^2$, whereas $|z|^2 = (a + bi)(a - bi) = a^2 + b^2$. Since $b \in \mathbb{R}$, $b^2 \geq 0$. Thus $a^2 + b^2 \geq a^2$. The equality holds when $b = 0$ (when z is real).

2.) Write the following expressions in the form $a + bi$, with $a, b \in \mathbb{R}$: (a) $\sqrt{i}$, (b) $(2 + i)^{2/3}$ (c) i^i .

Solution:

(a) $i = e^{i\pi/2} = \cos(\pi/2) + i \sin(\pi/2)$, so $\sqrt{i} = \left(e^{i\pi/2}\right)^{1/2} = e^{i\pi/4} = \cos(\pi/4) + i \sin(\pi/4) = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$. This is probably fine, but note that one could also write $i = e^{i(\frac{\pi}{2} + 2\pi n)}$ such that $\sqrt{i} = e^{i(\frac{\pi}{2} + 2\pi n)/2} = e^{i(\frac{\pi}{4} + \pi n)} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$ or $-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$. Note that even though we shifted the original complex number by full rotations in the complex plane, the roots are not trivial rotations of 2π from each other; they are unique solutions. This is better explained in problem 3.

(b) $(2 + i)$ has a magnitude of $\sqrt{5} = \sqrt{2^2 + (i)(-i)}$. We can write this in exponential form by finding an angle from the real axis: $\tan(\phi) = 1/2 \Rightarrow \phi = 26.6^\circ = 0.466$ rad. So $(2 + i) = \sqrt{5}e^{i\phi} \Rightarrow (2 + i)^{2/3} = (\sqrt{5}e^{i\phi})^{2/3} = 5^{1/3}e^{2i\phi/3} = 5^{1/3}e^{2i(0.466)/3} = 1.62 + 0.52i$.

Any of these forms is fine.

Once again, note that there are multiple solutions:

$$(2 + i) = \sqrt{5}e^{i\phi} = \sqrt{5}e^{i(\phi + 2\pi m)}$$

$$(2 + i)^{1/3} = 5^{1/6}e^{i(\phi/3 + 2\pi m/3)}$$

$$(2 + i)^{2/3} = 5^{2/6}e^{2i(\phi/3)}, 5^{2/6}e^{2i(\phi/3 + 2\pi/3)}, 5^{2/6}e^{2i(\phi/3 + 4\pi/3)} \\ = 1.63 + 0.52i, -0.36 - 1.67i, -1.26 + 1.15i$$

(c) We can write $i = e^{i\pi/2} \Rightarrow i^i = \left(e^{i\pi/2}\right)^i = e^{-\pi/2} = 0.208\dots$ which is real!

3.) Prove that the n^{th} root of a complex number z has n distinct solutions (that is, n roots that are not merely duplications of each other by additional rotations of multiples of 2π in the complex plane).

Solution:

If we write the complex number $z = Ae^{i\phi}$, then, more generally $z = Ae^{i(\phi + 2\pi m)}$ where m is any integer. The root we are asked for is $z^{1/n} = A^{1/n}e^{i\phi/n}$, but more generally $z^{1/n} = A^{1/n}e^{i(\phi/n + 2\pi m/n)}$. Although we arrived at this more general expression by shifting the imaginary exponent in the original representation of z by trivial factors of $2\pi m$, the roots that are possible are not trivial shifts of $2\pi m$. In the expression $z^{1/n} = A^{1/n}e^{i(\phi/n + 2\pi m/n)}$, we get different positions on the complex circle for different choices of $m < n$. Only when $m = n$ have we shifted the roots by a full 2π . So, we get distinct complex roots for $m = 0, m < n$ which is a total of n distinct roots.

Linear Algebra:

- 4.) Consider the space of all 2×2 matrices with complex entries, that is, matrices of the form

$$\begin{pmatrix} a_{11} + b_{11}i & a_{12} + b_{12}i \\ a_{21} + b_{21}i & a_{22} + b_{22}i \end{pmatrix}$$

Which of the following statements is/are correct concerning the dimensionality of the matrix?

- (a) The dimension is 2, because it can be written as a linear combination of two columns
- (b) The dimension is 2, because elements can be either real or complex
- (c) The dimension is 4, because there are two columns, and each can be real or complex
- (d) The dimension is 4, because it can be written as a linear combination of four orthogonal matrices
- (e) The dimension is 8, because it can be written as a linear combination of four orthogonal matrices, and each can be real or complex.

Solutions: Answer is (d) only. The reason will be made more clear in the solution to the next question. But the point of confusion is usually whether or not complex numbers add another factor of two to the dimensions. The resolution is that you have defined a complex vector space, which means that the coefficients can be complex. Since the coefficients can be complex, all that “complexity” ;) can be absorbed into the coefficients. The additional freedom is not in the vectors themselves, but in the coefficients.

- 5.) Convince any skeptics in the room of your answer to the previous question by writing the matrix

$\begin{pmatrix} 1 + 3i & 5 - 6i \\ 7 - 2i & 4 + 9i \end{pmatrix}$ in a form in which the dimensionality is obvious. That is, write it as a linear combination of obviously orthogonal vectors with complex coefficients.

Solution:

$$\begin{pmatrix} 1 + 3i & 5 - 6i \\ 7 - 2i & 4 + 9i \end{pmatrix} = (1 + 3i) \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + (5 - 6i) \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + (7 - 2i) \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + (4 + 9i) \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Hopefully it is clear that all these matrices are mutually orthogonal and that any “additional freedom” from having complex numbers is in the coefficients.

- 6.) Three friends (A, B, and C) are arguing about linear algebra, as friends do. Each claim that their set of vectors is complete (spans the space) and that the others’ do not. Help the friends reconcile their differences. The sets chosen by the friends are as follow:

Friend A: $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$

Friend B: $\left\{ \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 2 & 0 \\ 2 & 2 \end{pmatrix}, \begin{pmatrix} 3 & 3 \\ 0 & 3 \end{pmatrix}, \begin{pmatrix} 4 & 4 \\ 4 & 0 \end{pmatrix} \right\}$

Friend C: $\left\{ \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \right\}$

Solution:

Friend A is very clearly correct. One can argue this intuitively, since each matrix can be used to affect a single one of the four possible entries and no other. Alternatively, one can show that these are four mutually orthogonal matrices and that they must therefore span a four-dimensional space. To do that, use $\langle A|B \rangle = \text{Tr}(AB^T)$ to show (for example) that

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \text{ and } B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \text{ are orthogonal: } \langle A|B \rangle = \text{Tr}\left(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\right) = \text{Tr}\left(\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}\right) = 0.$$

Friend C is very clearly incorrect, since their second vector can be written as a linear combination of the third and fourth. They are not linearly independent and cannot span the space.

Friend B is one you might find tricky. The point is to note that you cannot produce any one vector by a linear combination of the other three (because of the positions of the zeros). Since we have four linearly independent vectors, they must span the space. We could use the Gram-Schmidt process to produce normalized orthogonal vectors from these (a basis).

- 7.) Clearly there may be many sets of vectors that are linearly independent. But is the set of mutually orthogonal, normalized, spanning vectors always unique (here “spanning” means the set is complete)? If so, explain why. If not, provide a counterexample.

Solution: No, it is not unique. One example is the set of 3 normalized, mutually orthogonal vectors in 3D space. The vectors $\hat{i} = (1,0,0)$, $\hat{j} = (0,1,0)$, $\hat{k} = (0,0,1)$ and $\hat{i} = (1,0,0)$, $\hat{j} = \frac{1}{\sqrt{2}}(0,1,1)$, $\hat{k} = \frac{1}{\sqrt{2}}(0,1,-1)$ both satisfy these criteria. This is trivial in this Cartesian picture. But it becomes important to realize that these “rotations” of vector spaces are totally OK when switching from the eigenbasis of one operator to another, or diagonalizing a degenerate subspace.

- 8.) Consider the following linear algebra question:

“Given a Hermitian $n \times n$ matrix M with n distinct eigenvalues, must its corresponding eigenvectors form a complete, orthogonal set?”

One could give a formal proof or counterexample, but it might not be worth the class time. Also, we are interested in using this matrix stuff in quantum mechanics. So, instead of a proof, argue whether or not the above statement should be true **IF** the eigenvalues correspond to real physical observables and the eigenvectors correspond to real physical states of some system.

Solution: If an operator corresponding to a real physical observable is given by an $n \times n$ Hermitian matrix, then its n distinct eigenvalues correspond to values of the physical quantity that could be measured in a lab (energy, momentum, spin, etc). If the eigenvalues are not equal, then the probability of a system having one measured observable m_1 and simultaneously having a different value m_2 for the same measured observable is clearly zero. Thus $\langle 1|2 \rangle = 0$ for two eigenvectors of the matrix. Since this must be true for any pair of the n eigenvectors, the eigenvectors form a set of n mutually orthogonal vectors, which must span an n dimensional space. Thus, they form a complete, orthogonal set.

9.) Check your claim from the previous problem to see if it is consistent with the matrix

$$M = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

That is, calculate the eigenvectors and show that they do or do not form a complete set of orthogonal vectors. Find the unitary matrix that diagonalizes M.

Solution: We start by finding the eigenvalues:

$$\begin{vmatrix} -\lambda & 1 & 0 \\ 1 & -\lambda & 1 \\ 0 & 1 & -\lambda \end{vmatrix} = -\lambda(\lambda^2 - 1) - 1(-\lambda) = 0 \Rightarrow -\lambda(\lambda^2 - 2) = 0 \\ \Rightarrow \lambda = 0, \pm\sqrt{2}$$

Now using each eigenvalue, we find

$$\lambda = 0: \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = 0 \begin{pmatrix} a \\ b \\ c \end{pmatrix} \Rightarrow b = 0, a + c = 0 \Rightarrow \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$\lambda = \sqrt{2}: \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \sqrt{2} \begin{pmatrix} a \\ b \\ c \end{pmatrix} \Rightarrow b = \sqrt{2}a = \sqrt{2}c, a + c = \sqrt{2}b \Rightarrow \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}$$

$$\lambda = -\sqrt{2}: \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = -\sqrt{2} \begin{pmatrix} a \\ b \\ c \end{pmatrix} \Rightarrow b = -\sqrt{2}a = -\sqrt{2}c, a + c = -\sqrt{2}b \Rightarrow \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix}$$

We can now check the orthogonality of the vectors:

$$(1, 0, -1) \begin{pmatrix} 1 \\ \pm\sqrt{2} \\ 1 \end{pmatrix} = 1 + 0 - 1 = 0 \\ (1, \sqrt{2}, 1) \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix} = 1 - 2 + 1 = 0$$

Three mutually orthogonal vectors clearly span the 3D space. If we really wanted to, we could make linear combinations of these to produce the familiar, trivial spanning vectors (1,0,0) etc. But that's overkill.

10.) Which of the following pairs show a vector space and a subspace within that vector space, and which do not?

- (a) The set of all n –tuples of real numbers, the set of all $(n - 1)$ –tuples of real numbers
- (b) The set of all n –tuples of complex numbers, the set of all n –tuples of real numbers. Careful here.
- (c) The set of all eigenstates of S^2 , the set of all eigenstates of S_z
- (d) The set of all functions of x , the set of all real-valued functions of x . Careful again.
- (e) The set of all functions of x satisfying $f(0) = 1$, the set of all functions of x satisfying both $f(0) = 1$ AND $f(1) = 0$
- (f) The set of all functions of x satisfying $f(0) = 0$, the set of all functions of x satisfying both $f(0) = 0$ AND $f(1) = 0$

Solution:

- (a) the set of all $n-1$ tuples is not contained within the set of all n -tuples. So, this is not a subspace within the described vector space.
- (b) It depends on whether we are working on a field of real or complex coefficients. If we have real coefficients only, then it works, since n -tuples of complex numbers form a vector space, and n -tuples of real numbers exist within that space and the subspace of n -tuples of real numbers is closed under addition. If we allow complex coefficients, then the subspace is not closed under addition.
- (c) The eigenstates of S^2 DO for a vector space, since S^2 is proportional to the identity... every vector is an eigenvector. However, the eigenstates of S_z do not form a subspace, since it is not closed under addition (a superposition of eigenstates is not an eigenstate, in general).
- (d) Depends on whether we allow real or complex coefficients.
- (e) Nope. The larger set isn't even a vector space since it is not closed under addition.
- (f) FINALLY, yes! This one describes a valid vector subspace within a larger vector space, regardless of allowed coefficients.

11.) Your friend claims that the set of functions $f_n = a_n \sin(b_n x)$ forms a complete orthonormal set in her Hilbert space. She's a pretty smart person, so you assume she is correct. Which of the following could be her Hilbert space? For any that are possible Hilbert space for your friend, describe the inner product.

- (a) The set of all single-valued functions of the (1-D) variable x
- (b) The set of all single-valued functions of the (1-D) variable x on the interval $0 \leq x < 2\pi$
- (c) The set of all single-valued functions of the (1-D) variable x on the interval $-\pi < x < \pi$ that take on a value of zero at $x = 0$
- (d) The functions describing spatial eigenstates of a 1-D infinite square well with walls at $x = 0$ and $x = L$
- (e) The functions describing spatial eigenstates of a 1-D infinite square well with walls at $x = -L/2$ and $x = L/2$

Solutions:

- (a) No, $f_n(x = 0) = 0$, so not all single-valued functions could be written as a linear combination of these.
- (b) No, same reason.
- (c) No, only odd functions on that interval can be generated with a linear combinations of f_n s.
- (d) Yes.
- (e) No, only odd functions.